On the Limitations of Cross-lingual Encoders as Exposed by Reference-Free Machine Translation Evaluation

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Text Generation Evaluation

Traditional Evaluation: compare system-generated text $\mathbf{y}$ with human written reference $\mathbf{y}^\star$.

- Require costly human references ☹
- Evaluation is limited to the parallel data available ☹

Reference-free Evaluation: compare system-generated text $\mathbf{y}$ with source text $\mathbf{x}$.

- No human references required ☺
- Allow for domain-agnostic (unlimited) evaluation ☺

Reference-free Evaluation

In text summarization, $\mathbf{x}$ and $\mathbf{y}$ with large length differences are written in the same language.

— Create a pesudo-reference $\tilde{\mathbf{x}}$ from $\mathbf{x}$ and compare $\tilde{\mathbf{x}}$ and $\mathbf{y}$ (Our other work at ACL2020[1])

In Machine Translation, $\mathbf{x}$ and $\mathbf{y}$ are short texts and written in different languages.

— Compare $\mathbf{x}$ and $\mathbf{y}$ directly in a cross-lingual space (this work)

[1] SUPERT: Towards New Frontiers in Unsupervised Evaluation Metrics for Multi-Document Summarization

Current State of Affairs

Unlike reference-based MT evaluation, there are almost no lexical overlaps in source and system-generated text.

Related work thus uses the following approaches to compare $\mathbf{y}$ with $\mathbf{x}$:

- a bag-of-word translation model [Popović, 2012]
- language-agnostic linguistic features [Specia et al., 2013]
- cross-lingual semantic frames [Lo et al., 2014]
- multilingual BERT embeddings and optionally integrates cross-lingual semantic roles and frames [Lo, 2019]

This Work: Reference-free MT Evaluation

- Can we use human labels for $\mathbf{y}$ and $\mathbf{y}^\star$ to judge the quality of reference-free metrics?

- We compare $\mathbf{x}$ and $\mathbf{y}$ using Multilingual BERT (M-BERT) and LASER, using the combination of:

  — a set-based or cosine distance metric

  — re-aligned contextualized embeddings to handle cross-lingual mismatch of mutual translations

  — a target-side LM to handle fluency

  $\mathbf{y}$: Disease case: when paying the organiser
  $\mathbf{x}$: Krankheitsfall: Wann bezahlt der veranstalter

Legitimacy of Reference-free Evaluation

Human judgments assign one direct assessment (DA) score to a pair of $\mathbf{y}$ and $\mathbf{y}^\star$, whereas $m(\mathbf{y}, \mathbf{x})$ compares $\mathbf{y}$ with $\mathbf{x}$.

We validate if DA scores are a valid proxy in the cross-lingual setup.

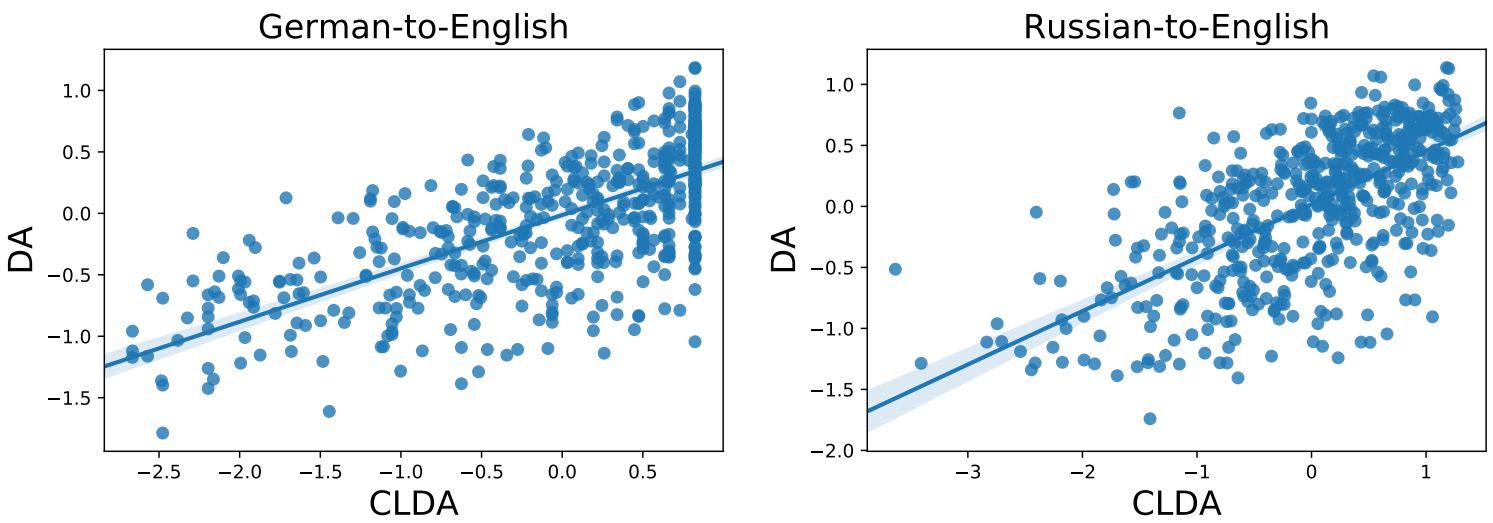


Figure: (Four) Annotators directly compare source and translation pairs. The small-scale manual analysis hints that DA scores are a valid proxy for CLDA.

A Cross-lingual Set-based Metric

The cross-lingual extension of MoverScore [Zhao et al., 2019].

- Input two sets of words $\mathbf{x}$ and $\mathbf{y}$, and a distance matrix $\mathbf{C}$
- Each $\mathbf{C}_{ij}$ in $\mathbf{C}$ has: $\|\text{M-BERT}(\mathbf{x}_i) - \text{M-BERT}(\mathbf{y}_j)\|_2$
- Optimization problem: find the optimal many-to-many word alignments $\mathbf{F}$ in two sets

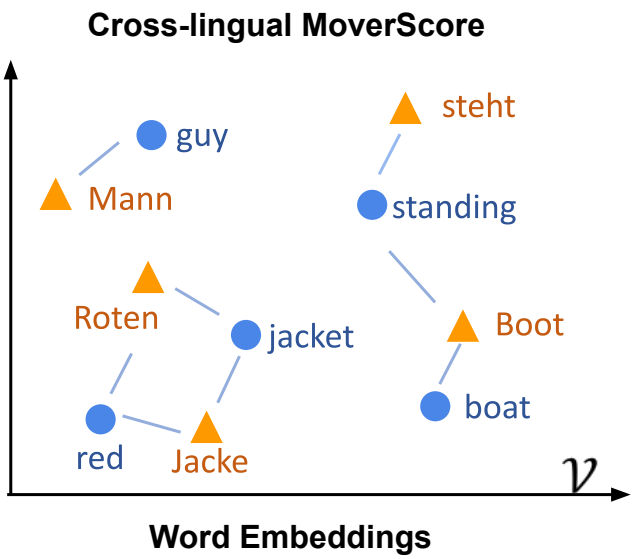
$$g = \min_{\mathbf{F} \in \mathbb{R}^{|\mathbf{x}^n| \times |\mathbf{y}^n|}} \sum_{ij} \mathbf{C}_{ij} \cdot \mathbf{F}_{ij}$$

- Normalize the total cost as a metric

$$m(\boldsymbol{x}, \boldsymbol{y}) := \frac{1}{1+g}$$

A Cross-lingual Set-based Metric

- Node: contextualized word embedding
- Edge: the cost of aligning two embeddings $0 \leq \mathbf{F}_{ij} \leq 1$



- x: A **guy** with a **red jacket** is **standing** on a **boat**
- y: **Ein Mann mit einer Roten Jacke steht auf einem Boot**

Re-aligned Contextualized Embeddings

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We'll illustrate with two post-hoc cross-lingual alignments:

- Linear Cross-lingual Projection (CLP)
- Universal Language Mismatch-Direction (UMD)

Linear Cross-lingual Projection

- Input set $\mathcal{D}$, two matrices $\mathbf{X}_\ell$ and $\mathbf{X}_k$
- Each $d \in \mathcal{D}$ is a pair of mutual word or sentence translations (w_l, w_k)
- Each vector in $\mathbf{X}_\ell$ and $\mathbf{X}_k$ is a pair of word/sentence embedding of w_l and w_k
- Optimization problem: find the rotation matrix $\mathbf{W}$ of two languages

$$\min_{\mathbf{W}} \|\mathbf{W}\mathbf{X}_\ell - \mathbf{X}_k\|_2.$$

- Procrustes Problem: a closed-form solution

$$\hat{\mathbf{W}} = \mathbf{U}\mathbf{V}^\intercal, \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\intercal = \text{SVD}(\mathbf{X}_\ell \mathbf{X}_k^\intercal)$$

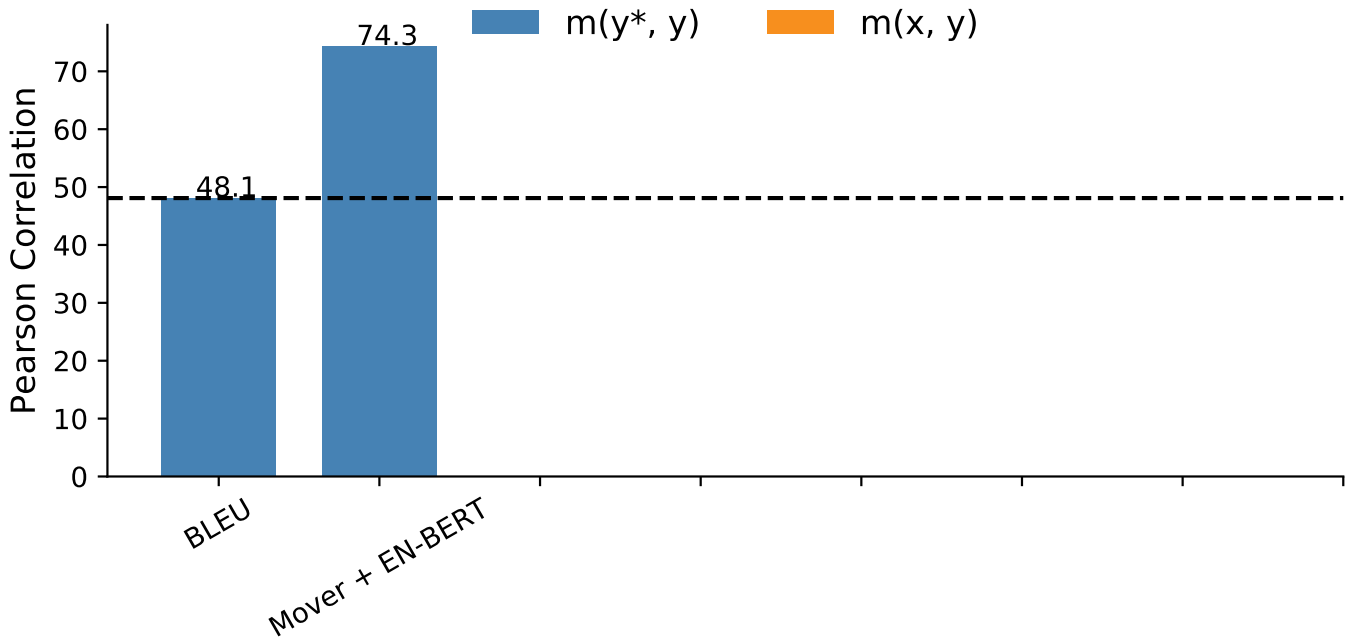
Universal Language Mismatch-Direction

- Input set $\mathcal{D}$ and a matrix $\mathbf{Q}$
- Each $d \in \mathcal{D}$ is a pair of mutual word or sentence translations (w_l, w_k)
- Each vector in $\mathbf{Q}$: $\text{Emb}(w_\ell) - \text{Emb}(w_k)$
  (a language-bias (misalignment) vector)
- Obtain the top singular vector v_B of $\mathbf{Q}$
  (the main direction of misalignment of two languages)
- Modify all embeddings

$$\tilde{\text{Emb}}(w_\ell) = \text{Emb}(w_\ell) - \underbrace{\cos(\text{Emb}(w_\ell), v_B) v_B}_{\text{its projection onto } v_B}$$

Results on WMT-2017 Dataset

Average results of metrics correlated with humans:



- Re-mapping has considerable effect ☺
- A target-side LM is required to outperform BLEU ☹
- The gap between $m(\mathbf{y}^\star, \mathbf{y})$ and $m(\mathbf{x}, \mathbf{y})$ is still big ☹

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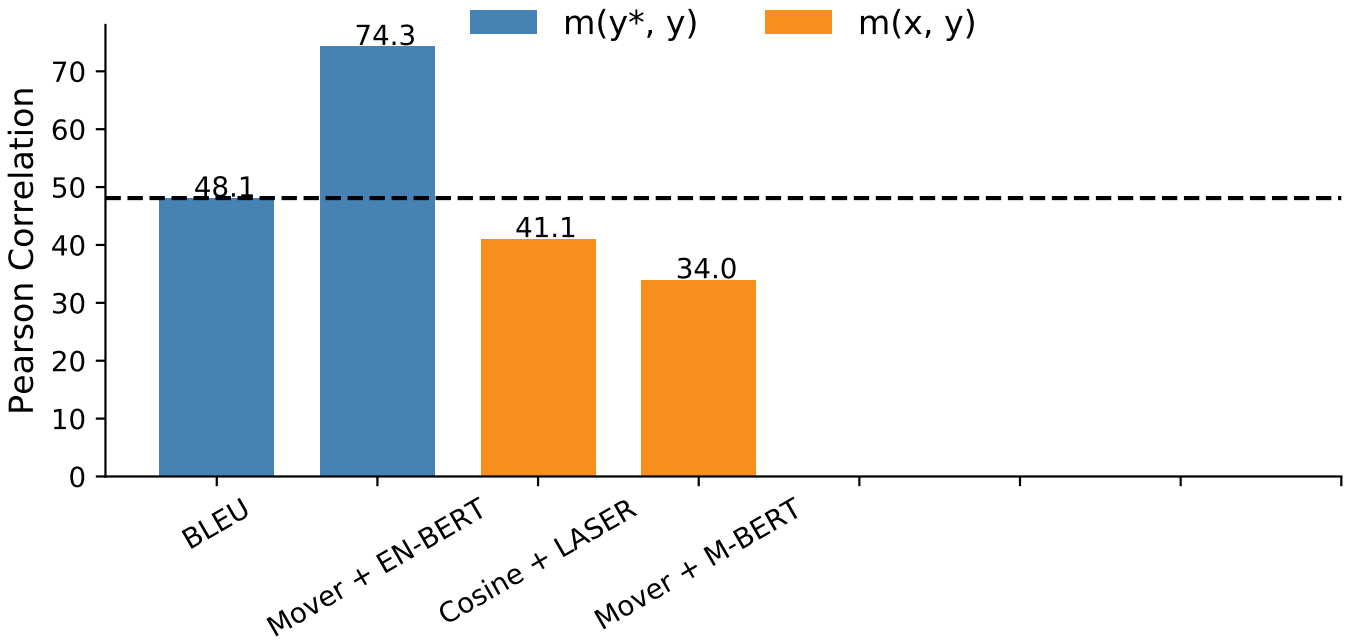

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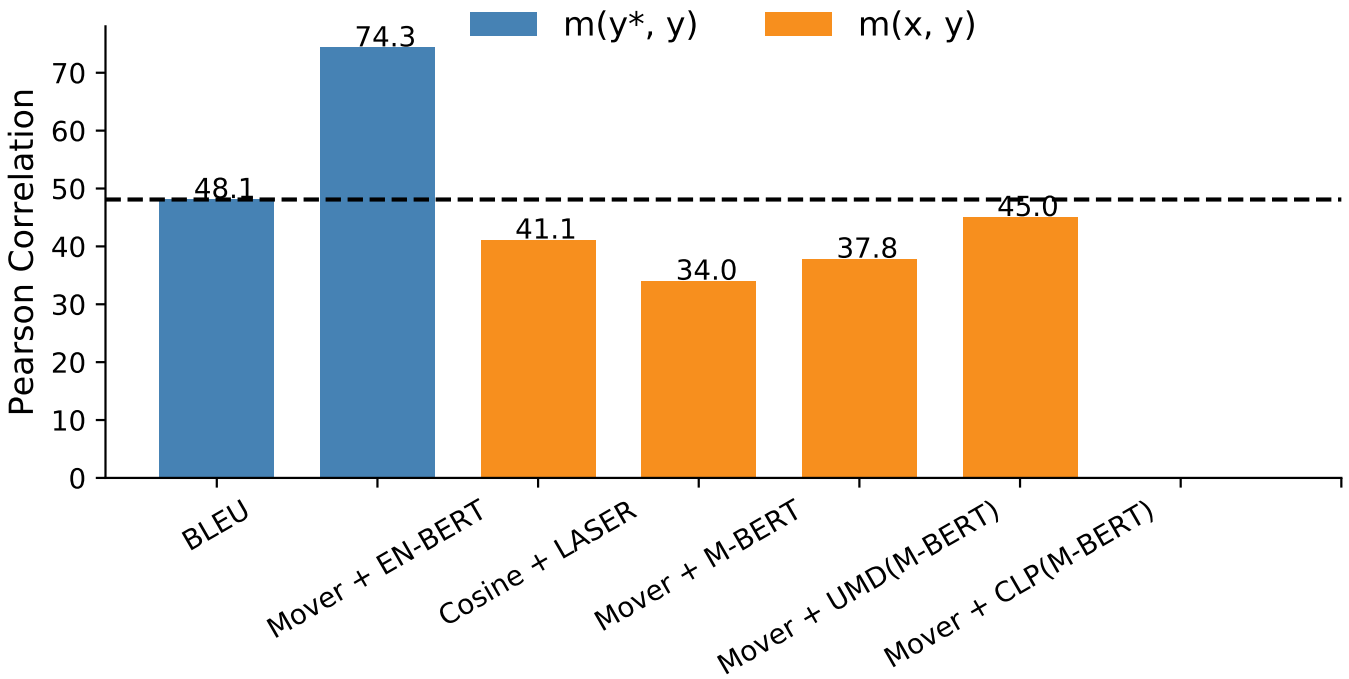

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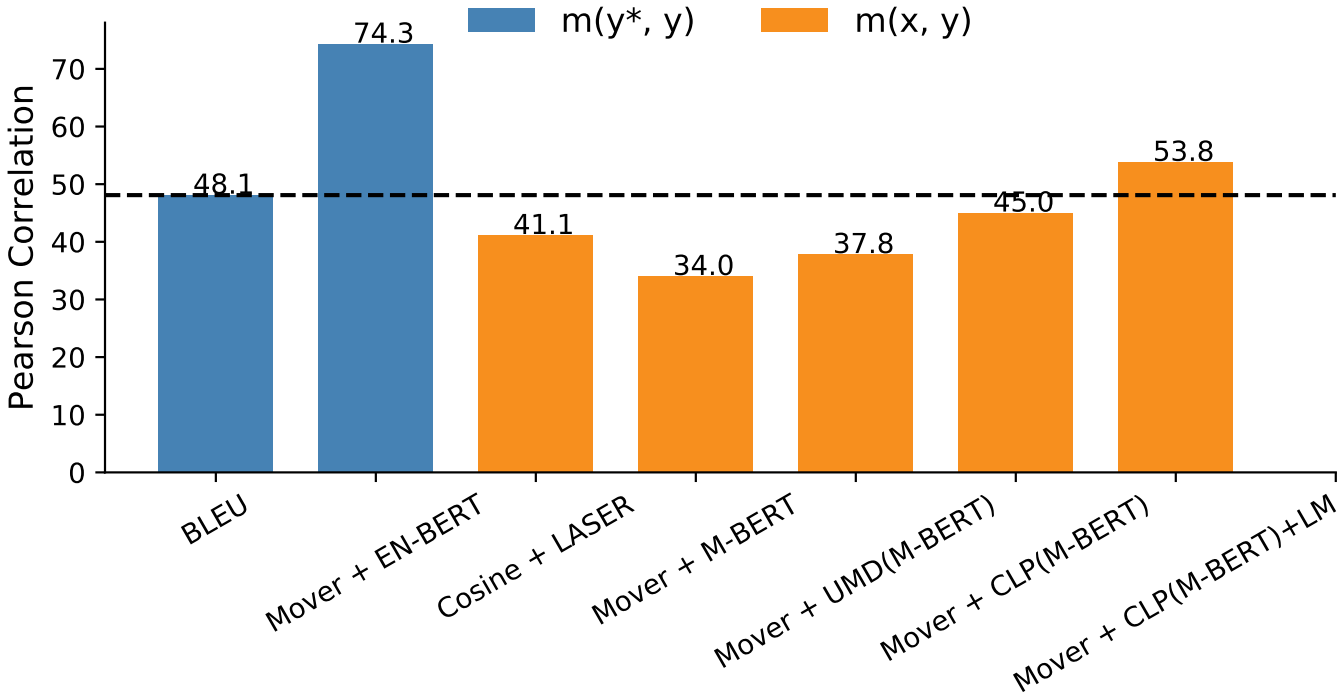

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Why do our metrics underperform?

- Evaluate the metric preference

$$d(\tilde{\mathbf{y}}, \mathbf{y}^\star; \mathbf{x}) := m(\mathbf{x}, \tilde{\mathbf{y}}) - m(\mathbf{x}, \mathbf{y}^\star)$$

When $d > 0$, then metric m prefers $\tilde{\mathbf{y}}$ over $\mathbf{y}^\star$. Here, let $\tilde{\mathbf{y}}$ be

- a random reordering of $\mathbf{y}^\star$, i.e. a bag-of-words sentence
- an expert reordering of $\mathbf{y}^\star$ to have the same word order as $\mathbf{x}$ i.e. translationese without semantic changes
- a word-by-word translation by human i.e. translationese with semantic changes

Illustration

$\mathbf{x}$	Dieser von Langsamkeit geprägte Lebensstil scheint aber ein Patentrezept für ein hohes Alter zu sein.
$\mathbf{y}^\star$	However, this slow pace of life seems to be the key to a long life.
$\mathbf{y}^\star$-random	To pace slow seems be the this life. life to a key however, of long
$\mathbf{y}^\star$-reordered	This slow pace of life seems however the key to a long life to be.
$\mathbf{x}'$-expert	This of slow pace characterized life style seems however a patent recipe for a high age to be.

Table: German source sentence $\mathbf{x}$, the human reference $\mathbf{y}^\star$, a randomly ($\mathbf{y}^\star$-random) and expertly reordered ($\mathbf{y}^\star$-reordered) English sentence, and literal translation ($\mathbf{x}'$) of $\mathbf{x}$ by human.

Breaking X-lingual Metrics with Translationese

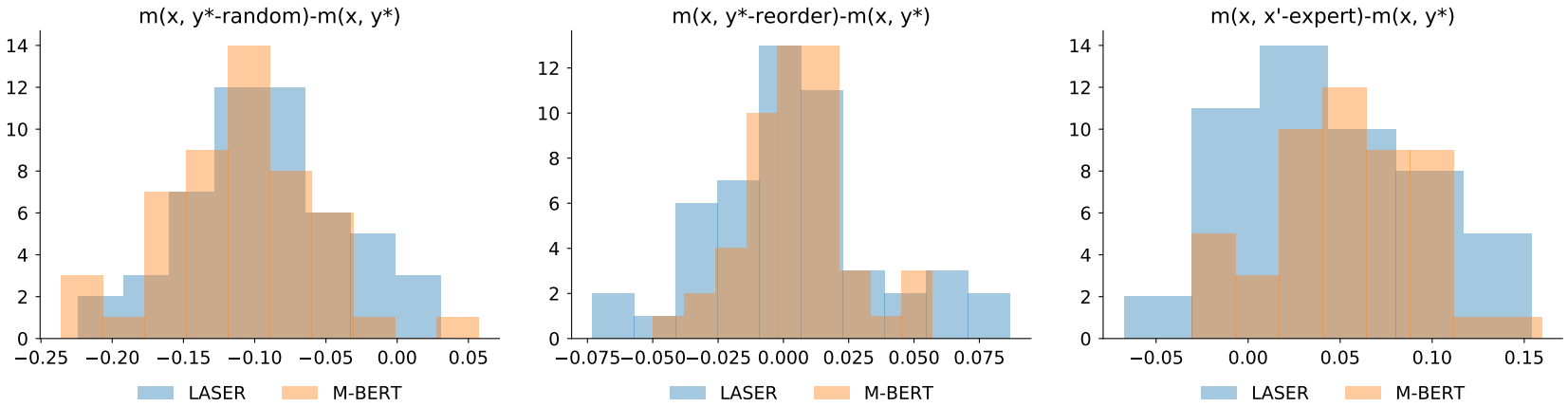
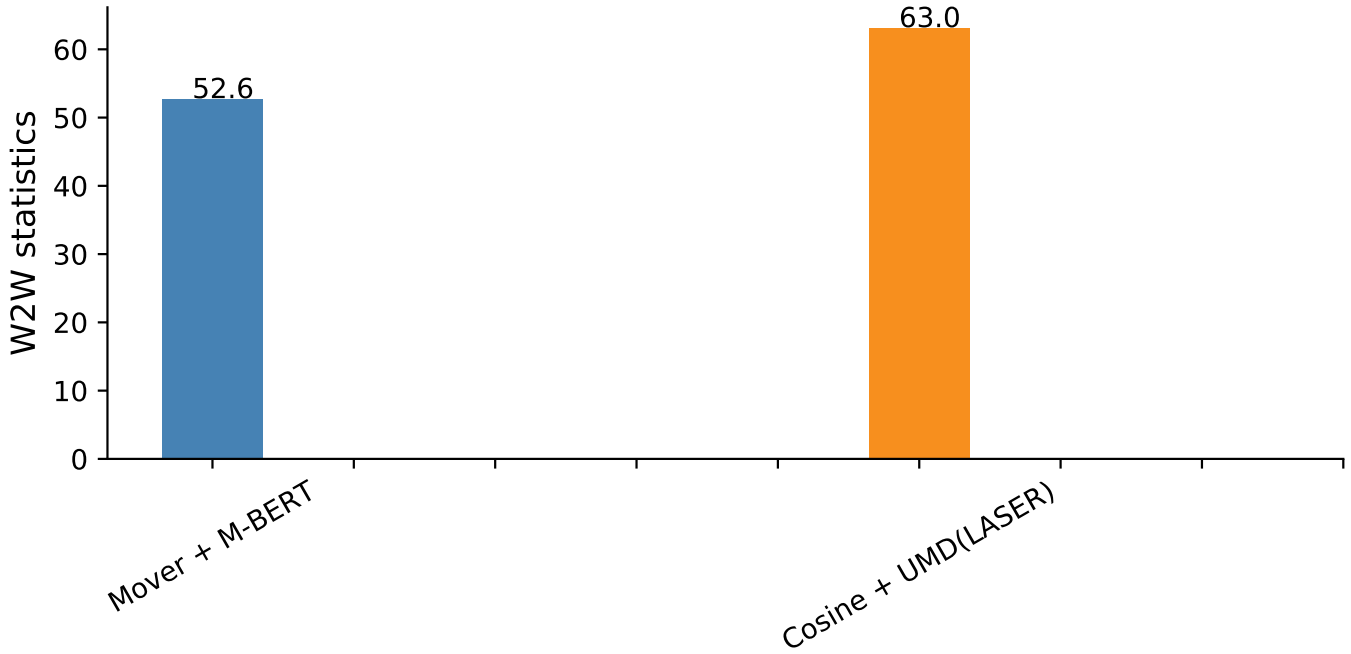


Figure: Histograms showing the metric preferences.

- Favor gold over randomly-shuffled human references ☺ (confirming they do not have bag-of-words property)
- Roughly indifferent between gold and reordered human references ☹
- Favor expert word-by-word translations over gold human references ☹ ☹

Breaking X-lingual Metrics with Translationese

Average results (lower is better) of the relative number of times that $m(\tilde{\mathbf{y}}, \mathbf{x})$ prefers literal over human translation.



- Cross-lingual re-mapping helps to a small degree ☹
- A target-side LM penalizes word-by-word translations ☺

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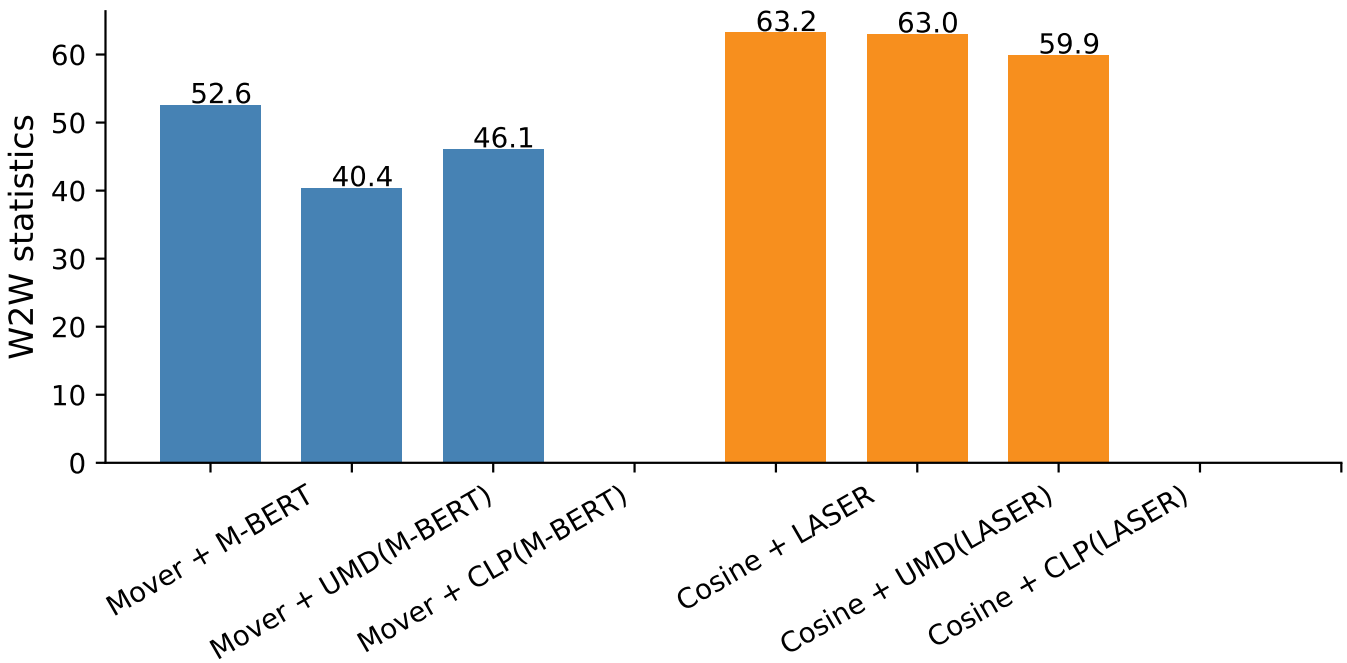

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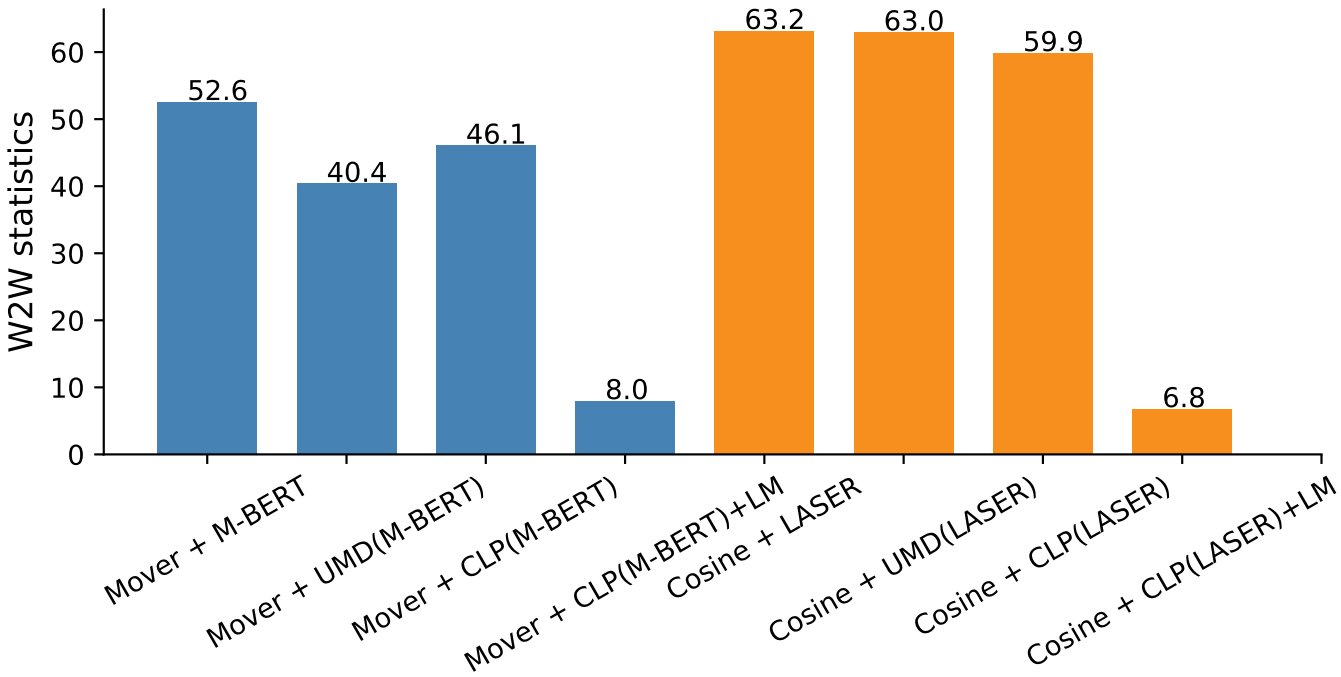

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Conclusion & Resource

- We identified important limitations of contextualized embeddings in natural adversarial setup—reference-free MT evaluation, suggesting that it should be included in future Xlingual benchmarks.
- Our metric, dubbed XMoverScore, surpassed the BLEU baseline, giving a viable prospect of wide-scale domain-agnostic MT evaluation.
- Release our metric at https://github.com/AIPHES/ACL20-Reference-Free-MT-Evaluation

Thänk yoü!

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